

MINYANG LI

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) *Sep. 2023 – Jun. 2027 (expected)*

B.E. in Artificial Intelligence

Current GPA: 3.414/4.3

Selected Coursework: Deep Learning (**A+**), Introduction to Artificial Intelligence (**A+**), Creative Coding and Interactive Art (**A+**), Introduction to Computer Vision (**A**), Introduction to Data Science and Analytics (**A**), Introduction to Computer Science (**A**), Computational Techniques for Sketch-based Creativity (**A**).

National University of Singapore (NUS)

Jun. 28 – Jul. 18, 2026

Incoming exchange Student, nominated by HKUST(GZ)

RESEARCH INTERESTS

My research interests center on computer vision and visual generative models, with a primary focus on diffusion models. I am also interested in their connections to embodied AI, including Vision-Language-Action (VLA) systems, diffusion models for decision making, and RL-based preference optimization for diffusion models and VLMs.

RESEARCH EXPERIENCE

Topic: High-Resolution Image Synthesis [1]

Apr. 2025 - Aug. 2025

Advisor: Prof. Ying-Cong Chen and Mr. Zhen Yang

ENVISON Lab at HKUST(GZ)

- Contributed to a training-free high-resolution synthesis method for pre-trained diffusion models, enabling image generation beyond the native training resolution without additional model training.
- Implemented and extended the method to multiple downstream generation tasks, including image editing, customized generation, controllable generation, and video generation.
- Took primary responsibility for extensive experimental validation of the method by benchmarking quantitative metrics and qualitative perceptual quality against prior baselines across image generation and these downstream tasks.
- Participated in rebuttal-stage technical analysis, clarifying the methodological differences between our approach and related prior work, particularly regarding sampling-trajectory truncation and guidance-based compensation.

Topic: Computational Hydrology [2]

Jun. 2024 - Sep. 2024

Advisor: Prof. Qichun Yang

Earth, Ocean and Atmospheric Sciences Thrust at HKUST(GZ)

- Adopted a deep learning framework for hydrological and geospatial time-series prediction.
- Built the preprocessing pipeline for large-scale spatiotemporal datasets, covering resolution alignment, cleaning, and training-ready formatting.
- Trained and evaluated LSTM-based models for precipitation prediction across China.

INTERNSHIPS

AI Researcher (Algorithm Intern)

Mentor: Dr. Tianyi Zhang

Mar. 2026 – Present

Knowin AI at Nanshan, Shenzhen, China

- Currently working as a core contributor in building a pose foundation model for language-conditioned keypose prediction rather than continuous action generation to improve embodiment transferability and compatibility with downstream IK, motion planning, and controllers.
- Adapted the VLA backbone to predict pick/place keyposes from observation images and task prompts.
- Designed a phase-wise VQ-VAE action tokenizer with separate pick/place codebooks, soft oracle targets, and phase-specific loss weighting.
- Established the robotic data collection SOP, led hands-on collection program, and supported real-world deployment of pick-and-place demo model.

PUBLICATIONS

- [1] Zhen Yang*, Guibao Shen*, **Minyang Li***, Liang Hou, Mushui Liu, Luo Zhou Wang, Xin Tao, and Ying-Cong Chen, “RectifiedHR: Enable efficient high-resolution synthesis via energy rectification,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*, Jun. 2026, pp. 3809–3819.
- [2] Haoyuan Yu, Qichun Yang, Liuqian Yu, Xia Li, **Minyang Li**, and Yingxian Yang, “Future climate change increases streamflow and risks of hydrological hazards in the Pearl River Basin,” *Water*, vol. 18, no. 3, p. 436, 2026.

SELECTED PROJECTS

Multi-Object Image Editing

Independent

Jun. 2025 - Jul. 2025

Course Project in Deep Learning

- Proposed VGFE, an inversion-free flow-based image editing framework that uses VQA-guided editing strength search to automatically select optimal edit strength while reducing computational overhead. Extended the method to multi-object editing by designing a reassembly and re-editing pipeline tailored for inversion-free diffusion/flow models. (Report)

Skeleton-Adhered Style Transfer for Chinese Characters

Collaborator: Jian Yang

Oct. 2025 - Dec. 2025

Course Project, in CompTec for Sketch-based Creativity

- Developed CalliGen, a sketch-based Chinese calligraphy generation system that transforms rough hand-written skeletons into stylized calligraphic characters while preserving users’ structural intent. Designed an efficient pipeline to construct a large-scale paired dataset of Chinese character images and skeletons for training. Co-designed an interactive prototype supporting sketch editing, style selection, and real-time generation (Report)(Code)(Demo video)

Fact-Aware Consistency Scoring for Model-Generated Answers

Collaborator: Zhiyi Chen, Zhiling Li

Apr. 2026 - May. 2026

Course Project in Intro2NLP

- To address the limitations of global embedding similarity in detecting entity, date, number, and span-level errors, we proposed FS-BGE, a BGE-M3 ColBERT-based similarity score with factual token reweighting and answer-span mismatch penalties, and I further explored bidirectional NLI-based entailment scoring and scoring in hyperbolic space. (Report)(Code)

SKILLS

**Programming, Web
AI, Robotics
Systems, Tools**

Python, basics of C++ and JavaScript, LaTeX
PyTorch, Hugging Face Diffusers, OpenCV, RL Bench, LeRobot
Linux HPC Management, NumPy, Pandas

*Equal contribution.